

Smart Grid Technology / Renewable Energies

Engineering Technology

May, 2026

Smart Grid Technology / Renewable Energies (A13361)

Short Title: Smart Grid Tech / Renew Engy
Faculty Science and Computing
Department Engineering Technology
Cluster Engineering
Author CLAIRE.FITZPATRICK
Credits 5

Level: Advanced

Description of Module / Aims

The student will learn that while the existing electrical power, transmission and distribution system has served the world well that a number of factors including the intermittency of the increasingly important renewable resources mean that a more agile, monitored and distributed grid, a smart grid, will form a part of their future. This course begins by highlighting the bountiful but unreliable nature of the dominant renewables in use on the electrical grid, wind, hydro and solar photovoltaic. The matching of supply and demand, as well as unyielding target of mains frequency together with the only slightly less rigid voltage targets, will then be covered. At this point, the challenge should be clear, so the final sections of the course deal with the solution and technology that would make up the smart grid. This begins with consumer monitoring and incentives through smart metering and demand side management, follows on with technology to automatically address grid problems and finally is completed with an overview of energy storage technologies. Practical work will involve building an automated measurement system based on LabVIEW and using the Smart Grid Project developed in the previous year of the course.

Programmes

ELEC-0044	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	4 / 8 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	4 / 8 / M
ELEC-0044	BEng (Hons) (WD_ETRIC_B)	4 / 8 / M
ELEC-0044	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 8 / E

Indicative Content

- Electric grid renewables
- Introduction to energy resources (wind, solar, hydro)
- Electrical energy fundamentals: voltage, frequency and supply/demand
- Challenges of distributed, renewable energy grid integration
- Smart electrical, smart metering and demand side management
- Smart grid sensing, measurement, communications, control & automation
- Energy storage

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the key parameters influencing the electrical power output from a renewable energy resources (Wind, Solar, hydro).
2. Compare the difficulties posed by integrating renewable energy into a grid with the integration of fossil fuel generators.
3. Evaluate the relationship between, and requirements imposed for, voltage and frequency in an electrical grid and fluctuations in supply/demand.
4. Design a smart electrical grid based on current and proposed future technology identifying each of the key components, their key parameters and scope and finally their relative impacts in addressing typical and atypical grid requirements.
5. Test some basic grid fundamentals on a smart grid demonstrator developed by the student, based on the Smart Grid Project completed in the previous year of this course.

Learning and Teaching Methods

- Lectures
- Tutorials
- Laboratory work

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	40%	1,2,3
<i>Project</i>	60%	4,5

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Practical	12	
Independent Learning	87	

Essential Material

- National Instruments Online. _Getting Started with NI LabView Student Training_ .. 2010.
- Sumper, A. and A. Baggini. _Electrical Energy Efficiency: Technologies and Applications_ .. Wiley, 2012.

Requested Resources

- ENGINEERING LAB: Electrical